

Note: Choose the best option and record your choice on the Answer Sheet following this part. If answers are not transferred to the sheet, they may not be graded.

PART-1 (20 Marks)

- If for a graph search problem, the cost of every action (move) is the same, breadth first search is guaranteed to return an optimal solution.
(A) True (B) False
- For a 3×3 tic-tac-toe game, there are _____ different paths (from start to the end of game) in its state space graph.
(A) 3^9 (B) 9^3 (C) $3!$ (D) None of the given options
- If symmetry of initial symbol placement is considered, the above tic-tac-toe game has _____ paths through its state space graph.
(A) $3 \times 7!$ (B) $12 \times 7!$ (C) $8 \times 7!$ (D) None of the given options
- Which of the following is true about data-driven search?
(A) It is also called backward chaining (B) It starts at the goal node(s)
(C) It involves searching from facts to the goal(s) (D) It is feasible when goal(s) of a problem are known.
- For a shallower search tree, depth first iterative deepening search compared with depth first search, in general, is:
(A) More efficient (B) Equally well (C) Optimal (D) Inefficient
- For the 8-puzzle problem, the state space has _____ states accessible from any random start state.
(A) $8!$ (B) $9!$ (C) $9!/2$ (D) None of the given options
- $\alpha\beta$ search is always better than the minimax search.
(A) True (B) False
- α cut-off means:
(A) Prune further possibilities below the current MAX node
(B) Prune further possibilities below the current MIN node
- If d represents depth and b , the average branching factor, the worst-case space complexity of $\alpha\beta$ search can be characterized as:
(A) $O(b^d)$ (B) $O(b \log d)$ (C) $O(d^b)$ (D) $O(bd)$
- Admissibility of a heuristic always implies:
(A) Informedness (B) Monotonicity (C) Irrevocability (D) None of the given options
- In best first search, which of the following is true:
(A) A set of sorted paths is added to the queue
(B) Entire queue is sorted each time a new path is added
(C) Chronological backtracking is done to switch to promising paths
(D) All the nodes with a specified heuristic value(s) are explored first
- _____ is a conceptual representation, while _____ is a concrete one.
(A) Search Tree, State Space (B) State Space, Search Tree
- Consider a search tree with depth $d = 3$ and branching factor $b = 2$, the number of nodes explored by depth first iterative deepening search in worst case would be:
(A) 8 (B) 15 (C) 26 (D) 22

$$\frac{1-2^4}{1-2^3}$$

$$= \frac{-15}{-5}$$

odd

$$1-b^{d+1}$$



8
7
15

14. For certain problems, e.g. fractional knapsack problem, hill climbing is always guaranteed to give optimal solution.
(A) True (B) False
15. Heuristic evaluation function for the uniform cost search may be characterized as:
(A) $f(n) = g(n) + h(n)$; where $h(n) = 0$ (C) $f(n) = g(n) + h(n)$; where both $g(n)$ and $h(n)$ are non-negative
(B) $f(n) = g(n) + h(n)$; where $g(n) = 0$ (D) None of the given options
16. Admissibility is a stricter requirement than monotonicity.
(A) True (B) False
17. TSP optimization problem is an NP hard problem.
(A) True (B) False
18. A larger tournament size would result in a more diverse offspring population.
(A) True (B) False
19. Genetic algorithm can be characterized as a powerful variant of _____.
(A) Uniform cost search (B) Breadth first search (C) Hill climbing search (D) None of the given options
20. Mutation helps ensure that _____ is present in the offspring population.
(A) Stagnation (B) Diversity (C) Inversion (D) None of the given options

Answer Sheet

Question	Choice
1	B X
2	C X
3	C X
4	C ✓
5	B X
6	B X
7	A ✓
8	B ✓
9	A X
10	D ✓
11	B ✓
12	B ✓
13	C ✓
14	B X
15	A ✓
16	A X
17	A ✓
18	A X
19	C ✓
20	B ✓

11

Answer Sheet

Question	Choice
1	A ✓
2	D ✓
3	C X
4	D X
5	D ✓
6	C ✓
7	B X
8	B ✓
9	A X
10	D ✓
11	B ✓
12	B ✓
13	C ✓
14	B X
15	A ✓
16	B ✓
17	A ✓
18	B ✓
19	C ✓
20	B ✓

15

PART-2 (50 Marks)

Note: Attempt all questions.

Q.1. [1+2+4+3]

A farmer wants to move himself, a fox, a goose and some grain across a river. Unfortunately, his boat is so tiny that he can take only one of his possessions across the river on any trip and he is the one who could operate the boat. Worse yet, an unattended fox will eat a goose, and an unattended goose will eat grain! So, the farmer must not leave the fox alone with the goose or the goose alone with the grain. If the farmer and his possessions are all on the left bank of the river, what sequence of moves must the farmer follow to get to the right bank?

Consider representing the above problem using the state space representation as follows.

Using the 4-tuple representation for a problem state as $\langle a_1, a_2, a_3, a_4 \rangle$, where the indices 1 through 4 represent the farmer, the fox, the goose, and the grain, respectively, and $a_i \in \{L, R\}$, where L and R designate left and right banks of the river, do the following.

(a) What would the start and goal states look like?

L	R
a_1	
a_2	
a_3	
a_4	

start state : $\langle a_1, a_2, a_3, a_4 \rangle$ where $a_i \in \{L\}$

goal state : $\langle a_1, a_2, a_3, a_4 \rangle$ where $a_i \in \{R\}$

L	R
	a_1
	a_2
	a_3
	a_4

(b) Give the states that must be avoided given the constraints of the problem.

states avoided according to given constraints:

~~$\langle a_2, a_3 \rangle$~~

* $\langle a_1, a_2, a_3, a_4 \rangle$ where $a_1 \in \{R\}$ and $(a_2, a_3) \in \{L\}$

* $\langle a_1, a_2, a_3, a_4 \rangle$ where $a_1 \in \{R\}$ and $(a_3, a_4) \in \{L\}$

* $\langle a_1, a_2, a_3, a_4 \rangle$ where $a_1 \in \{L\}$ and $(a_2, a_3) \in \{R\}$

* $\langle a_1, a_2, a_3, a_4 \rangle$ where $a_1 \in \{L\}$ and $(a_3, a_4) \in \{R\}$

(c) Draw the state space graph showing only the permissible moves.

$$\langle a_1, a_2, a_3, a_4 \rangle ; a_i \in \{L\}$$



$$\langle a_1, a_2, a_3, a_4 \rangle ; (a_1, a_3) \in \{R\} \ \& \ (a_2, a_4) \in \{L\}$$



$$\langle a_1, a_2, a_3, a_4 \rangle ; (a_1, a_3)$$

$$\langle a_1, a_2, a_3, a_4 \rangle$$

avoided
not
pe

$$a_i \in \{L\}$$

a_1 farmer
 a_2 fox
 a_3 goose
 a_4 grain
 a goose fox

$$(a_1, a_2) \in \{R\}$$

$$(a_1, a_3) \in \{R\}$$

$$(a_3, a_4) \in \{L\}$$

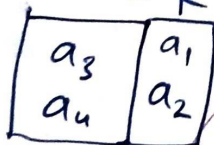
$$(a_2, a_4)$$

L R

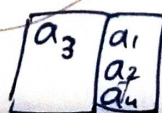
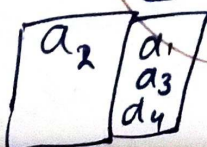
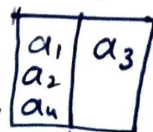
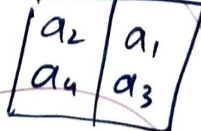


X

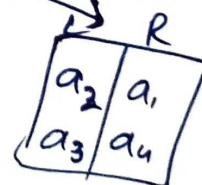
L R



L R



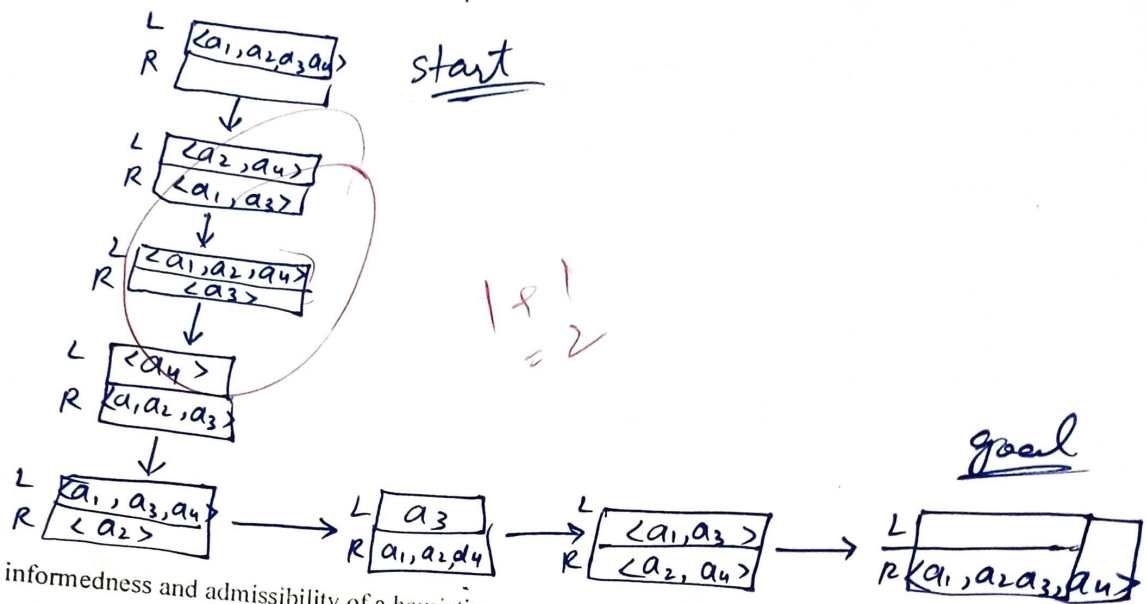
X



farmer
 fox
 fox

goose
 farmer
 fox

(d) Give the sequence of moves to reach the goal state. Use the tuple notation for a state and arrows to show transition between the states.



Q.2. [4+6]

(a) Differentiate between informedness and admissibility of a heuristic.

* Informedness tells us that from two heuristics, which heuristic gives less underestimate is more informed.

If $h_1(n) \leq h_2(n)$ then h_2 is more informed.

* Admissibility of a heuristic states that a heuristic always underestimates the cost to reach goal than actual cost.

(b) Show that monotonicity implies admissibility.

$$h(n) \leq h^*(n)$$

It always gives shortest path.

Monotonicity implies admissibility. It means that a monotonic heuristic is always admissible.

Let for some node n_i & n_j where j is descendent of i .
then

$$h(n_i) - h(n_j) \leq h(n_i, n_j)$$

where $h(n_i, n_j)$ is the cost to reach from n_i to n_j .
This shows that a monotonic heuristic always underestimates the cost to reach the goal than actual cost which shows that monotonic heuristic is always admissible.

Q.3. [2+4+4]

2

Consider the pseudo code for the best first search algorithm and answer the following questions.

```

function best_first_search;
begin
    open := [Start]           % initialize
    closed := []
    while open ≠ [] do       % states remain
        begin
            remove the leftmost state from open, call it X
            if X = goal then return the path from Start to X
            else
                begin
                    generate children of X
                    for each child of X do
                        case
                            the child is not on open or closed:
                                begin
                                    assign the child a heuristic value
                                    add the child to open
                                end
                            the child is already on open
                                if the child was reached by a shorter path
                                    then give the state on open the shorter path
                            the child is already on closed
                                if the child was reached by a shorter path then
                                    begin
                                        remove the state from closed
                                        add the child to open
                                    end
                        end
                    end
                end
                % case statements end
            put X on closed
            re-order states on open by heuristic merit % (best leftmost)
        end
    end
    return FAIL % open is empty
end

```

(a) Does the pseudo code above work fine? Give your answer as a 'Yes' or 'No'.

No, we should remove base on the

(b) Change the pseudo code if you answer 'No' in (a) above. Note: Highlight the section you think is not correct and give the modified statement(s) below.

remove the node which is nearest to the goal state
on basis of heuristic → which ~~is~~ less underestimates the goal state

(c) Modify the given pseudo code if the heuristic you use is consistent (monotone).

2

(a) How does Alpha-Beta pruning help improve the performance of the Minimax algorithm? Give the total game nodes examined by the Alpha-Beta search in the best as well as the worst-case scenario.

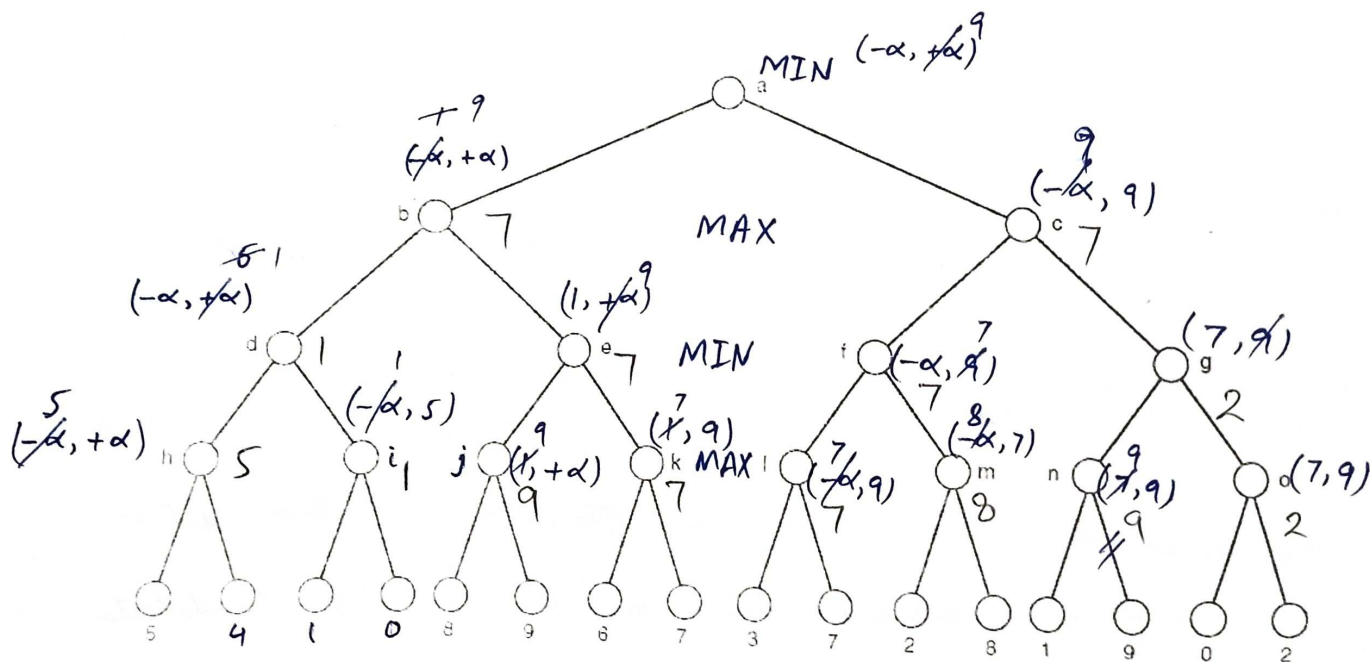
In alpha-Beta pruning, we prune the paths, we do not even generate those nodes which do not take part as deciding factor of path. This reduces the memory consumption to a large extent. 03

For Alpha-Beta Search:

In Best Case $\Rightarrow O(b^{d/2}) \approx O(b^{1/2})^d = O(\sqrt{b})^d$

In worst Case $\Rightarrow O(b^d)$

(b) For the search tree shown below it is MIN's turn. You are required to compute the final values of the pair (α, β) for all non-leaf nodes generated by the algorithm and fill them in the given table. What move does the MIN select?
Note: Do the conventional left to right depth-wise traversal. You must NOT evaluate (α, β) at the leaf nodes, which implies that the propagation of (α, β) pair must be stopped one level above the ply depth!



min will select b node
because there will be an ~~alpha cut-off~~ at ~~c node~~.
alpha cut-off at a towards c.

Node	Final (α, β)
a	$(-\alpha, +\alpha)$
b	$(-\alpha, +\alpha)$ $(9, +\alpha)$
c	$(9, 9)$
d	$(-\alpha, +\alpha)$ $(-\alpha, 1)$
e	$(1, 9)$
f	$(-\alpha, 7)$
g	$(7, 9)$
h	$(-\alpha, +\alpha)$ $(5, +\alpha)$
i	$(1, 5)$
j	$(9, +\alpha)$
k	$(7, 9)$
l	$(7, 9)$
m	$(8, 7)$
n	$(9, 9)$
o	$(7, 9)$

Q.5. [2+6+2]

05

(a) Consider bit-level representation for traveling salesperson problem involving 'n' cities. Describe one issue each in designing the crossover and mutation operators for this representation.

In

* ~~Cross over~~ the cross-over, ~~the~~ some of the cities may be repeated when cities in the candidate solutions are not ordered. Therefore, here we will use ordered cross-over.

* In mutation, we will eliminate the cities, but we require all the cities. Therefore this will not work

(b) For the following candidate solutions, apply order-crossover as discussed in the class to create two new solutions C_1 and C_2 . You must take the cut points as follows: first between city index 3 and city index 4, second between city index 7 and city index 8.
 Note: The candidate solutions shown below do not include the starting city. The list indexing starts from '1' NOT '0'

$$P_1 = (1, 9, 2, 4, 6, 5, 7, 8, 3)$$

$$P_2 = (4, 5, 9, 1, 8, 7, 6, 2, 3)$$

~~$$C_1 = (3, 8, 7, 5, 6, 4, 2, 9, 1)$$~~

$$C_1 = (5, 9, 2, 4, 1, 8, 7, 6, 3)$$

~~$$C_2 = (4, 5, 9, 1, 8, 7, 6, 2, 3)$$~~

~~$$C_2 = (4, 5, 9, 1, 8, 7, 6, 2, 3)$$~~

$$C_2 = (4, 5, 9, 1, 6, 2, 7, 8, 3)$$

(c) How does tournament selection work in genetic algorithm? What happens if the tournament size is too large?

In genetic algorithm, we do tournament selection.

Here we use ordered cross-over to and fix cut-paths to save some order of the genes in the candidate solutions.

* When tournament size is too large, it generate a diverse offspring population.